

Instructor: Dr. Jim D. Pham
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Office Hrs: Thursday, 7 PM – 8PM (Canvas)
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Section: CS/CNET-146 (Online)

Course Description:

This course provides students with the knowledge and skills necessary to be comfortable working in a UNIX or Linux environment. This course is taught using a combination of lectures, demonstrations, discussions, and hands-on labs. This course aligns to the Linux Professional Institute (LPI) Linux Essentials Professional Development Certificate.

Prerequisites:

Advisory: Math 151, 153 (Algebra I) or equivalent, and knowledge of the Windows operating system environment.

Textbook (Optional): Linux Essentials 2nd Edition

Author: Christine Bresnahan and Richard Blum
ISBN: 9781119092063
Copyright Year: 2015
Publisher: John Wiley & Sons, Incorporated

Student Learning Outcomes

After completing the course, students will be able to:

1. Demonstrate understanding of command line syntax, standard input and standard output, redirection, pipes, and background processing.
2. Demonstrate use of the command line to create and manage files and directories: find, copy, move, remove, and rename files; change file permissions
3. Demonstrate use of the vi text editor to create and access files, delete and insert text, search and replace text.
4. Create simple shell scripts.

Teaching Methods:

1. Lectures: The presentations slides and assignments will be accessed through Ohlone's Canvas Course Management System. See Instructional Resources below.
2. Assignments: End of chapter activities and online activities will be assigned weekly to reinforce material in the text. These assignments may require the application of various software packages.
3. Quizzes: Weekly unannounced quizzes may be given to help ensure students keep up with assigned material.
4. Exams: Three exams will be given: a) Midterm; b) Final Exam; and c) Comprehensive Exam. The exams will be delivered online, they will be timed, and will test assigned readings and material discussed in class. The final exam will not be comprehensive in nature. However, the instructor reserves the right to retest on material that was not appropriately comprehended.
5. Participation: Student participation will be graded by the level of class participation on the Discussion Board. Each student will be required to do 2 posts per chapter on the Discussion Board.

Instructional Resources:

1. **NetSpace** – Cisco Network Academy Site online learning environment (the textbook) - provides access to the Linux Essentials course content. <https://www.netacad.com> You will need a separate account for this site. If have completed any of the Cisco Networking courses you may use that same account.
2. **Canvas:** Class documents, syllabus, discussion, and the lecture slides will be posted on Ohlone's Canvas. Canvas is an online Course Management System. You access Canvas at this URL: <https://ohlone.instructure.com>.

Courseware:

NDG Linux Essentials, web-based course provides an introduction to Linux. The course includes curriculum content, lab exercises, and assessments within an engaging, hands-on learning environment. For an overview of the course and course content use the following link: http://www.netdevgroup.com/content/lpi/labs/lpi_linux_essentials.html. The NDG Linux Essentials is accessible through the Cisco Network Academy NetSpace.

Email:

If you send me email please put **course number – name – topic** into the email **Subject** field.
For example: CS/CNET 146 – Name – Issue

Labs and Assignments:

A Lab will be assigned each week starting with Module 4. Students will be expected to use the virtual machine provided in the Linux Essentials courseware available on NetSpace.

Exams and Tests:

There will be Chapter Exams, a Midterm Objective Test, a Final Objective Test and a comprehensive Final Exam. You will have three attempts for each Chapter Exam and single attempts for the Midterm and Final(s). For all exams you will be allowed to use of books, notes, or a Linux/UNIX command line. The questions will be multiple-choice, true/false, and short answer fill-ins. All exams will be timed however.

Policy on Student Work:

Labs and assignments are to be done individually, unless a group project is specified. You are encouraged to talk to fellow students about concepts and methods, but you may not copy someone else's work. I adhere to the **Ohlone College Policy on Academic Integrity**.

GRADING:

Canvas Participation:	10%
Chapter Exams:	10%
Labs:	20%
Midterm:	20%
Final Objective Test:	20%
Final Comprehensive Exam	20%

GRADING SCALE:

A = 90-100
B = 80-89
C = 65-79
D = 55-64
F = <55

Course Schedule: The following is a recommended schedule. The instructor reserves the right to make any schedule changes deemed necessary.

Session	Date	Activity
1	8/28/17	Module 1: The Computing Environment, Introduction to Linux - Chapter 1 Exam - Due date: 9/05/17
2	9/04/17	Holiday: Labor Day (Ohlone College closed)
3	9/06/17	Modules 2 and 3: Open Source Applications and Licenses , Using Linux - Chapter 2 Exam - Due date: 9/12/17
4	9/12/17	Modules 4 and 5: Command Line Skills, Getting Help - Chapters 4 and 5 Exam - NetSpace Labs 4 and 5 - Due date: 9/19/17
5	9/19/17	Module 6: The Linux File System (Managing Files), Working with Files and Directories - Chapter 6 Exam - NetSpace Lab 6 - Due date: 9/26/17
6	9/26/17	Module 7: Maintaining and Administering a Linux System and Backups, Archiving and Compression - Chapter 7 Exam - NetSpace Lab 7 - Due date: 10/03/2016
7	10/03/17	Module 8: The Shell – the Command Line and Regular Expressions, Pipes, Redirection and REGEX - Chapter 8 Exam - NetSpace Lab 8 - Due date: 10/10/2016
8	10/10/17	Midterm
9	10/17/17	Module 9: Introduction to Shell Scripts, Basic Scripting - Chapter 9 Exam - NetSpace Lab 9 - Due date: 10/24/17
10	10/24/17	Module 10: Understanding Computer Hardware - Chapter 10 Exam - NetSpace Lab 10 - Due date: 10/31/17
11	10/31/17	Module 11: Managing Packages and Processes NetSpace Lab 11 - Chapter 11 Exam - NetSpace Lab 11 - Due date: 11/07/17
12	11/07/17	Module 12: Network Configuration - Chapter 12 Exam - NetSpace Lab 12 Due date: 11/14/17

13	11/14/17	Modules 13 and 14: System and User Security and Create a New User NetSpace Labs 13 and 14 - Chapters 13 and 14 Exam - NetSpace Labs 13 and 14 - Due date: 9/26/17
14	11/21/17	Module 15: Ownership and Permissions - Chapter 15 Exam - NetSpace Lab 15 - Due date: 9/26/17
15	11/28/17	Holiday: Thanksgiving (Ohlone College closed)
16	12/05/17	Module 16: Securing Permissions, Links and File Locations - Chapter 16 Exam - NetSpace Lab 16 - Due date: 9/26/17
17	12/12/17	Final Exam